

N57

Consider the steady planar motion of a liquid of density ρ , viscosity μ , thermal conductivity k , and specific heat c that flows down over an inclined plane of angle α due to the action of gravity, forming a layer of constant thickness h . To solve the problem, use the coordinate system (x, y) indicated in the figure, in which gravity can be expressed as $\vec{g} = g \sin(\alpha) \vec{e}_x - g \cos(\alpha) \vec{e}_y$. To analyze the resulting liquid flow, assume that the velocity components $v_x(y)$ and $v_y(y)$ are a function of the distance to the wall, and that the air surrounding the liquid has a density $\rho_a \ll \rho$ and a viscosity $\mu_a \ll \mu$, such that in the first approximation it moves without significant pressure differences, exerting on the liquid a negligible viscous force.

- Using continuity, determine the velocity component $v_y(y)$.

Obtain the pressure field $p(x, y)$ by integration of the y component of the momentum equation, demonstrating in particular that $\partial p / \partial x = 0$.

Compute $v_x(y)$.

Calculate the force per unit surface that the liquid exerts on the wall.

Determine the temperature in the liquid $T(y)$. In the calculation, assume that the thermal conductivity of air is very small compared with that of the liquid ($k_a \ll k$) and that the wall is at constant uniform temperature T_w .

Obtain the conductive heat flux from the liquid to the wall $(\vec{q} \cdot \vec{n})_w$, where $\vec{n}$ is appropriately oriented.

To analyze the energy balances emerging in the problem, consider the fixed control volume V_0 of unity length indicated in the figure.

- Compute the total energy dissipated per unit time inside the control volume $\int_{V_0} \phi_v dV$.
- Calculate the work per unit time done by gravity $\int_{V_0} \rho \vec{v} \cdot \vec{g} dV$.
- Use conservation of energy in the control volume to explain why $(\vec{q} \cdot \vec{n})_w = \int_{V_0} \phi_v dV = \int_{V_0} \rho \vec{v} \cdot \vec{g} dV$.

1) $\nabla \cdot \vec{v} = 0 \Rightarrow \frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} = 0 \Rightarrow \boxed{v_y = C = 0}$

2) $0 = -\frac{\partial p}{\partial y} - \rho g \cos \alpha \Rightarrow \boxed{p = -\rho g \cos \alpha y + f(x) = \rho g \cos \alpha (h - y) + p_a}$

3) $0 = \mu \frac{\partial^2 v_x}{\partial y^2} + \rho g \sin \alpha \Rightarrow \boxed{v_x = -\frac{\rho g \sin \alpha}{2\mu} y^2 + C_1 y + C_2}$

Boundary conditions: $y=0, v_x=0$; $y=h, \frac{\partial v_x}{\partial y} = 0$

4) $-\vec{p} \cdot \vec{n} + \vec{\tau} \cdot \vec{n} = -(p_a + \rho g \cos \alpha h) \vec{e}_y + \rho g \sin \alpha h \vec{e}_x$

5) $0 = k \frac{\partial^2 T}{\partial y^2} + \phi_v$

6) $\vec{q} \cdot \vec{n} = k \frac{\partial T}{\partial y} \Big|_{y=0} = \frac{(\rho g \sin \alpha)^2 h^3}{\mu}$

7) $\int_{V_0} \phi_v dV = \frac{(\rho g \sin \alpha)^2 \sin^2 \alpha}{\mu} \int_0^h (h^2 - 2hy + y^2) dy = \frac{(\rho g \sin \alpha)^2 \sin^2 \alpha}{\mu} \frac{h^3}{3}$

8) $\int_{V_0} \rho \vec{v} \cdot \vec{g} dV = \int_0^h \rho v_x g_x dy = \rho g \sin \alpha \int_0^h \frac{\rho g \sin \alpha}{\mu} \left(\frac{h^2}{2} y - \frac{y^3}{3} \right) dy = \frac{(\rho g \sin \alpha)^2 \sin^2 \alpha}{\mu} \frac{h^3}{3}$

9) $\frac{d}{dt} \int_{V_0} \rho \frac{v^2}{2} dV + \int_{\partial V_0} \rho \vec{v} \cdot \vec{n} ds = \int_{\partial V_0} \vec{q} \cdot \vec{n} ds + \int_{V_0} \rho \vec{v} \cdot \vec{g} dV - \int_{V_0} \phi_v dV \Rightarrow 0 = -\dot{Q}_w + \int_{V_0} \rho \vec{v} \cdot \vec{g} dV - \int_{V_0} \phi_v dV$

10) $\frac{d}{dt} \int_{V_0} \rho \frac{v^2}{2} dV + \int_{\partial V_0} \rho \vec{v} \cdot \vec{n} ds = -\int_{\partial V_0} \vec{q} \cdot \vec{n} ds + \int_{V_0} \rho \vec{v} \cdot \vec{g} dV - \int_{V_0} \phi_v dV \Rightarrow 0 = \int_{V_0} \rho \vec{v} \cdot \vec{g} dV - \int_{V_0} \phi_v dV$

11) $\vec{q} \cdot \vec{n} = k \frac{\partial T}{\partial y} \Big|_{y=0} = \frac{(\rho g \sin \alpha)^2 h^3}{\mu}$

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